Matias Agelvis

Software Engineer · Bordeaux, France

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SUMMARY

Software engineer and physicist specializing in Python backend development, API design, data processing, and optimization. Combines production backend experience — FastAPI, PostgreSQL, Docker, RabbitMQ — with a strong foundation in mathematical modeling and discrete optimization.

EXPERIENCE

Margaux Systems — Operations Research Engineer

Mar 2026 – Present

- Developing optimization and decision-support solutions for manufacturing scheduling and production planning.
- Building mathematical optimization models and analytical tools using Python and MiniZinc.
- Designing simulations, analyses, and software workflows to translate operational constraints into practical solutions.

SNAPP' — Python Backend Engineer (Contractor)

Feb 2025 – Mar 2026

- Developed and maintained production backend services for a banking project using Python/FastAPI.
- Designed asynchronous REST APIs, authentication/authorization middleware, and services communicating via RabbitMQ.
- Built multi-database persistence layers with PostgreSQL and SQLAlchemy.
- Developed and maintained automated tests with pytest, as well as integration tests.
- Redesigned a FIX 4.2 integration over TCP, replacing all existing code: message handling, sessions, reconnection, subscriptions, caching, and integration testing.
- Developed data-processing and optimization algorithms with a focus on performance, robustness, and reliability.

SIAL — Laboratory Technician

Mar 2024 – Mar 2025

- Conducted laboratory testing and sample preparation under established procedures and safety requirements.
- Managed experimental data, records, logistics, and equipment.

Independent Software Development

Oct 2022 – Mar 2024

- Developed independent software projects while pursuing software engineering opportunities.
- Built and deployed web applications and software tools.

Frontend Developer — Remote

Mar 2022 – Oct 2022

- Developed responsive web application features using React, Next.js, TypeScript, and Chakra UI.
- Implemented reusable UI components and worked on cross-browser compatibility and performance.

EDUCATION

Universidad de los Andes — B.Sc. Physics (cum laude)

Mérida, Venezuela

Background in nonlinear dynamics, chaos theory, mathematical modeling, and computational methods.

SKILLS

Languages: Python, SQL, JavaScript

Backend: FastAPI, asynchronous Python, REST APIs, RabbitMQ

Optimization: MiniZinc, discrete optimization, mathematical modeling, algorithms

SELECTED PROJECTS

OmegaInt — Arbitrary-precision integer library in C++

CellSV — High-performance XML-to-CSV conversion tool

Significado Total — Linguistics research application built with Python and Svelte

Kumigi — Web-based avatar designer

Warkdown-98 — Windows 98-themed markdown editor running in the browser, built with React and Vite